

Question Number	Answer	Additional Guidance	Mark
3a	there is no (significant) difference between the (mean) diameter of the zone of inhibition between samples from village B and village A	Three parts to this answer: can be expressed in many different ways	1

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3bi	• suitable table format with data (1) • correct column headings with units (1) • means correctly calculated to same number of places as data table (1)	<table><tr><td colspan="3">{Diameter / length} of zone of inhibition /mm</td></tr><tr><td>Village A</td><td>Village B</td></tr><tr><td>15.9</td><td>17.2</td></tr><tr><td>16.2</td><td>17.0</td></tr><tr><td>14.5</td><td>17.3</td></tr><tr><td>17.1</td><td>15.8</td></tr><tr><td>15.3</td><td>17.3</td></tr><tr><td>15.4</td><td>18.9</td></tr><tr><td>16.2</td><td>19.1</td></tr><tr><td>16.8</td><td>19.3</td></tr><tr><td>14.4</td><td>16.9</td></tr><tr><td>16.8</td><td>14.8</td></tr><tr><td>18.2</td><td>16.7</td></tr><tr><td>16.1</td><td>17.4</td></tr><tr><td>16.9</td><td>17.8</td></tr><tr><td>16.6</td><td>17.4</td></tr><tr><td>16.4</td><td>16.8</td></tr></table>			{Diameter / length} of zone of inhibition /mm			Village A	Village B	15.9	17.2	16.2	17.0	14.5	17.3	17.1	15.8	15.3	17.3	15.4	18.9	16.2	19.1	16.8	19.3	14.4	16.9	16.8	14.8	18.2	16.7	16.1	17.4	16.9	17.8	16.6	17.4	16.4	16.8	3
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3bii	- bar graph with linear scale, correct axis labels including units (1) - means plotted correctly (1) - range bars plotted correctly (1)	Mean diameter of zone of inhibition (mm) and bars labelled A and B Accept broken scale 16.2 and 17.3 ALLOW ecf from 3bi A 14.4 to 18.2 B 14.8 to 19.3	3

Question Number	Answer	Additional Guidance	Mark
3biii	- correct calculation of numerator (1) - correct substitution of given $\frac{(S_A)^2}{n}$ and $\frac{(S_B)^2}{n}$ (1) - correct answer (1)	Example of calculation $17.3 - 16.2 = (-) 1.1$ Accept $17.31 - 16.19 = 1.12$ $\frac{0.99}{15} + \frac{1.39}{15}$ $t = 2.76$ Accept $- 2.76 / 2.81 / - 2.81$ Allow mp 2 and 3 only if incorrect means used correct answer with no working gains (3) marks	3

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3biv	An answer that includes the following: - t value greater than critical value so {difference is significant / reject the null hypothesis} (1) - Bacteria from village A are (significantly) more resistant to ampicillin than bacteria in village B (1) - Cannot comment on development of resistance in village B as do not have any data to compare {before ampicillin given / with another village where ampicillin has not been given} (1)	Accept correct numbers $2.76 > 2.05$ Ref to correlation & difference = no mark Accept appears that resistance has developed in village A (as ZOI is significantly bigger) Accept that bacteria in village B have not developed resistance compared to bacteria in village A	2

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3c	An answer that includes two of the following: • more samples from each village (1) • compare with villages not using {antibiotic / ampicillin} (1) • control of biotic factors (1)	Accept repeat the study / larger sample size / samples from more villages eg {age / variety / species / diet / spacing} of chickens Ignore reference to genetic variation in chickens / mass / gender Ignore reference to testing different concentrations of	